

Tutorial Week 6

Question 1

User Login System FMEA

You're a quality engineer analyzing a simple user login system for a web application. The system authenticates users via a username and password. You need to perform an FMEA to identify potential failures and prioritize risks.

1. Identify two failure modes for the login system.
2. For each failure mode:
 - Describe the **effect** on the system or user.
 - Suggest a possible **cause**.
 - Assign **Severity (S)** (1-10: 1 = minor, 10 = catastrophic), **Occurrence (O)** (1-10: 1 = rare, 10 = frequent), and **Detection (D)** (1-10: 1 = always detected, 10 = undetectable).
 - Calculate the **Risk Priority Number (RPN)** = $S \times O \times D$.

Answer

Failure Mode	Effect	Cause	Severity	Occurrence	Detection	RPN
Incorrect password check	User cannot login	Logical error in code	7	3	4	84
Server timeout	Login hang	Network overload	5	4	3	60

Question 2

File Upload Feature FMEA

You're assessing a file upload feature in a cloud storage app. Users upload files via a web interface, and the system saves them to a server. Perform an FMEA to evaluate risks.

1. Identify two failure modes for the file upload feature.
2. For each failure mode:
 - Describe the **effect**.
 - Suggest a possible **cause**.
 - Assign **S**, **O**, and **D** ratings (1-10 scale).
 - Calculate the **RPN**.

Answer

Failure Mode	Effect	Cause	Severity	Occurrence	Detection	RPN
File corruption	User loss data	Transfer interrupt	9	3	4	108
Upload timeout	Upload failed, user retries	Slow server response	6	5	3	90

Question 3

Real-Time Video Streaming Service FMEA

You're a quality engineer for a real-time video streaming service (e.g., a live sports app). The critical function under analysis is “**Stream video to user device**”, which handles encoding, transmission, and playback. The system processes 10,000 streams/hour during peak events. Perform an advanced FMEA with quantitative RPNs based on the following data:

- **Historical Logs:**
 - 50 stream interruptions in 1,000 hours (0.05/hour/stream).
 - 20 buffering delays in 1,000 hours (0.02/hour/stream).
 - 10 decode errors in 1,000 hours (0.01/hour/stream).
 - 5 crashes in 1,000 hours (0.005/hour/stream).
 - **Detection Metrics:**
 - Monitoring catches 90% of interruptions, 80% of delays, 70% of decode errors, 50% of crashes.
 - **Impact Assessment:**
 - Interruption: 10% user dropout (severe).
 - Delay: 5% user dropout (moderate).
 - Decode error: 8% user dropout (significant).
 - Crash: 15% user dropout + complaints (critical).
1. Identify four failure modes for “Stream video to user device.”
 2. For each failure mode:
 - Describe the **effect** on users/system.
 - Identify a **cause**.
 - Assign quantitative **Severity (S)**, **Occurrence (O)**, and **Detection (D)** ratings (1-10 scale):
 - S: Based on user dropout % and impact (1 = <1%, 10 = >10%).
 - O: From failure rate/hour (1 = <0.001, 10 = >0.1).
 - D: From detection % (1 = >95%, 10 = <10%).
 - Calculate **RPN** = $S \times O \times D$.

Answer

Failure Mode	Effect	Cause	S	O	D	RPN
stream interruptions	10% user dropout	Network package	10	5	2	100

		loss				
buffering delays	5% user dropout	Insufficient bandwidth	6	3	3	54
decode errors	8% user dropout	Codec error/incompatible	8	2	4	64
crashes	15% user dropout + complaints	Memory overflow	10	1	6	60

Question 4

E-Commerce Order Processing Failure RCA

You're a reliability engineer performing an RCA for an e-commerce platform after a major outage where **“Order processing fails”**—customers couldn't complete purchases during a peak sale, costing \$50K/hour. The system handles order validation, payment, and inventory updates. Using FTA, analyze the failure based on:

- **Historical Data:**
 - Network disconnect: 0.02/hour (2% chance/hour).
 - Server overload (caused by both):
 - “High load” (0.05/hour)
 - “No scaling” (0.15, 15% chance of no auto-scaling).
 - Payment gateway failure: 0.03/hour (3% chance/hour).
 - Database lock (caused by both):
 - “Concurrency bug” (0.01/hour)
 - “No lock timeout” (0.2, 20% chance of missing timeout logic).
- **System Insight:** Any one of these—network issue, server overload, payment failure, or DB lock—can halt order processing (top-level OR gate).

You are required to: -

1. Construct the fault tree.
2. Calculate the probability per hour of “Order processing fails” (round to three decimal places).
3. Calculate MTTF.

Answer

- **Server Overload (AND):**
 - $P = 0.05 \times 0.15 = 0.0075/\text{hour}.$
- **DB Lock (AND):**

- $P = 0.01 \times 0.2 = 0.002/\text{hour}$.
- **Network:** 0.02/hour (direct).
- **Payment:** 0.03/hour (direct).
- **Top Event (OR):**
 - $P = 1 - (1 - 0.02) \times (1 - 0.0075) \times (1 - 0.03) \times (1 - 0.002)$.
 - $= 1 - 0.98 \times 0.9925 \times 0.97 \times 0.998$.
 - $= 1 - 0.9419438 = 0.0580562 \approx 0.058/\text{hour}$ (5.8% chance/hour).

$$\text{MTTF} = 1/0.058 = 17.24 \text{ hours}$$

Question 5

Cloud Backup System Failure RCA

You're investigating a cloud backup system failure where “**Backup fails to complete**”, leaving customer data unprotected during a critical window. The RCA aims to trace root causes after 10% of backups failed last month (100 failures in 1,000 hours). Data includes:

- **Network Issue (caused by either):**
 - “Packet loss” (0.04/hour)
 - “Slow bandwidth” (0.03/hour).
- **Storage Failure (caused by both):**
 - “Disk full” (0.02/hour)
 - “No auto-expansion” (0.25, 25% chance of no expansion logic).
- **Software Bug (caused by either):**
 - “Compression error” (0.015/hour)
 - “File lock” (0.01/hour)
 - “No retry” (0.3, 30% chance of no retry logic).
- **Top-Level Logic:** Any one of Network, Storage, or Software failure causes the backup to fail.

Your tasks: -

1. Construct the fault tree (provided above).
2. Calculate the probability per hour of “Backup fails to complete” (round to three decimal places).
3. Calculate MTTF.

Answer

- **Network Issue (OR):**

- $P = 1 - (1 - 0.04) \times (1 - 0.03) = 1 - 0.96 \times 0.97 = 1 - 0.9312 = 0.0688/\text{hour}.$

- **Storage Failure (AND):**

- $P = 0.02 \times 0.25 = 0.005/\text{hour}.$

- **Software Bug:**

- Lock Issue (AND): $P = 0.01 \times 0.3 = 0.003/\text{hour}.$

- Total (OR): $P = 1 - (1 - 0.015) \times (1 - 0.003) = 1 - 0.985 \times 0.997 = 1 - 0.982045 = 0.017955/\text{hour}.$

- **Top Event (OR):**

- $P = 1 - (1 - 0.0688) \times (1 - 0.005) \times (1 - 0.017955).$

- $= 1 - 0.9312 \times 0.995 \times 0.982045.$

- $= 1 - 0.909698 = 0.090302 \approx 0.090/\text{hour} (9.0\% \text{ chance/hour}).$